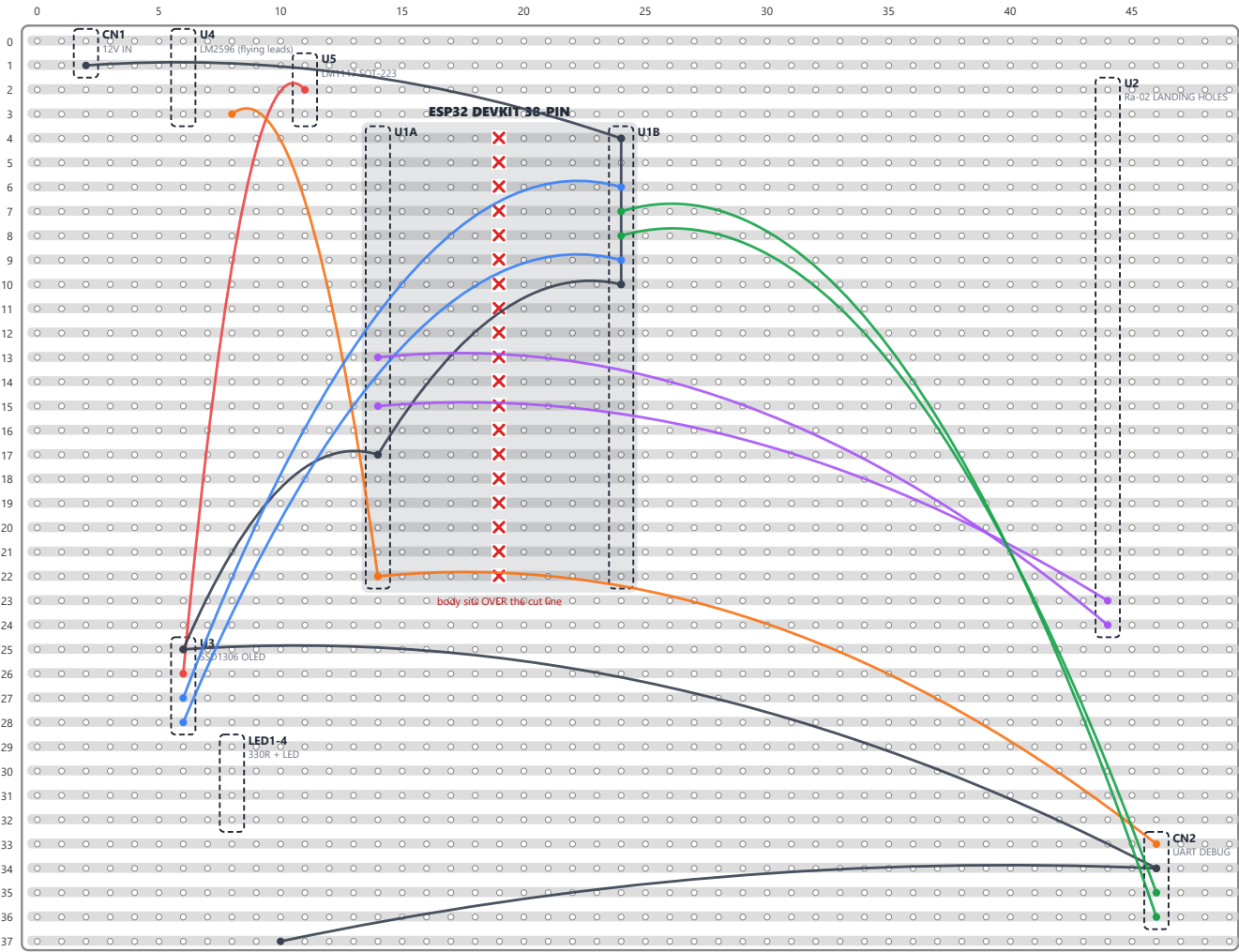


BS-02 Base Station

Veroboard build book — 19 strip cuts, 15 wire links. Strips run left-right.



BS-02 BASE STATION
19 strip cuts (red X) . 15 wire links . strips run left-right

Hole addresses are (col,row); the numbers run along the top and left edge of the diagram.

Step 1 — cut 19 strips

DO THIS BEFORE ANY PART GOES IN. Once the devkit is fitted you cannot reach the cuts underneath it. Meter every cut: probe either side, it must read open circuit.

☐	Row	Cut between	Separates
☐	4	col 14 and col 24	spare U1A.1 GND
☐	5	col 14 and col 24	spare U1A.2 SPI_MOSI
☐	6	col 14 and col 24	spare U1A.3 I2C_SCL
☐	7	col 14 and col 24	spare U1A.4 TX0
☐	8	col 14 and col 24	spare U1A.5 RX0
☐	9	col 14 and col 24	spare U1A.6 I2C_SDA
☐	10	col 14 and col 24	spare U1A.7 GND
☐	11	col 14 and col 24	spare U1A.8 SPI_MISO
☐	12	col 14 and col 24	spare U1A.9 SPI_SCK
☐	13	col 14 and col 24	LORA_DIO0 LORA_NSS
☐	14	col 14 and col 24	spare U1A.11 spare U1B.11
☐	15	col 14 and col 24	LORA_RST spare U1B.12
☐	16	col 14 and col 24	spare U1A.13 LED_B
☐	17	col 14 and col 24	GND spare U1B.14
☐	18	col 14 and col 24	LED_R LED_G
☐	19	col 14 and col 24	spare U1A.16 LED_Y
☐	20	col 14 and col 24	spare U1A.17 spare U1B.17
☐	21	col 14 and col 24	spare U1A.18 spare U1B.18
☐	22	col 14 and col 24	+5V spare U1B.19

Step 2 — strip map

CHECK THIS BEFORE SOLDERING INTO ANY NEW HOLE. Find the row, find the column range: that is what the hole is already joined to. Rows shaded pink must stay empty.

Row	Cols	Net	Pins in this segment
0	0-49	+12V	CN1.1@c2 U4.IN+@c6
1	0-49	GND	CN1.2@c2 U4.IN-@c6 U4.OUT-@c8 U5.1@c11 C5.1@c12 C1.1@c13 C2.1@c20
2	0-49	+3V3	U5.2@c11 C1.2@c13 C2.2@c20 U2.VDD@c44
3	0-49	+5V	U4.OUT+@c8 U5.3@c11 C5.2@c12
4	0-19	LEAVE EMPTY — devkit 3V3	U1A.1@c14
4	20-49	GND	U1B.1@c24
5	0-19	LEAVE EMPTY — devkit EN	U1A.2@c14
5	20-49	SPI_MOSI	U1B.2@c24 U2.MOSI@c44
6	0-19	LEAVE EMPTY — devkit VP/I036	U1A.3@c14
6	20-49	I2C_SCL	U1B.3@c24
7	0-19	LEAVE EMPTY — devkit VN/I039	U1A.4@c14
7	20-49	TX0	U1B.4@c24
8	0-19	LEAVE EMPTY — devkit I034	U1A.5@c14
8	20-49	RX0	U1B.5@c24
9	0-19	LEAVE EMPTY — devkit I035	U1A.6@c14
9	20-49	I2C_SDA	U1B.6@c24
10	0-19	LEAVE EMPTY — devkit I032	U1A.7@c14
10	20-49	GND	U1B.7@c24
11	0-19	LEAVE EMPTY — devkit I033	U1A.8@c14
11	20-49	SPI_MISO	U1B.8@c24 U2.MISO@c44
12	0-19	LEAVE EMPTY — devkit I025	U1A.9@c14
12	20-49	SPI_SCK	U1B.9@c24 U2.SCK@c44
13	0-19	LORA_DIO0	U1A.10@c14
13	20-49	LORA_NSS	U1B.10@c24 U2.NSS@c44
14	0-19	LEAVE EMPTY — devkit I027	U1A.11@c14
14	20-49	LEAVE EMPTY — devkit I017	U1B.11@c24
15	0-19	LORA_RST	U1A.12@c14
15	20-49	LEAVE EMPTY — devkit I016	U1B.12@c24
16	0-19	LEAVE EMPTY — devkit I012	U1A.13@c14
16	20-49	LED_B	U1B.13@c24 R.K2@c34
17	0-19	GND	U1A.14@c14
17	20-49	LEAVE EMPTY — devkit I00	U1B.14@c24
18	0-19	LED_R	R.K4@c8 U1A.15@c14
18	20-49	LED_G	U1B.15@c24 R.K1@c30
19	0-19	LEAVE EMPTY — devkit SD2	U1A.16@c14
19	20-49	LED_Y	U1B.16@c24 R.K3@c38
20	0-19	LEAVE EMPTY — devkit SD3	U1A.17@c14
20	20-49	LEAVE EMPTY — devkit SD1	U1B.17@c24
21	0-19	LEAVE EMPTY — devkit CMD	U1A.18@c14
21	20-49	LEAVE EMPTY — devkit SD0	U1B.18@c24
22	0-19	+5V	U1A.19@c14
22	20-49	LEAVE EMPTY — devkit CLK	U1B.19@c24
23	0-49	LORA_RST	U2.RST@c44
24	0-49	LORA_DIO0	U2.DIO0@c44
25	0-49	GND	U3.1@c6
26	0-49	+3V3	U3.2@c6

27	0-49	I2C_SCL	U3.3@c6
28	0-49	I2C_SDA	U3.4@c6
29	0-49	K1	R.K1@c30 LED.K1@c32
30	0-49	K2	R.K2@c34 LED.K2@c36
31	0-49	K3	R.K3@c38 LED.K3@c40
32	0-49	K4	R.K4@c8 LED.K4@c10
33	0-49	+5V	CN2.1@c46
34	0-49	GND	CN2.2@c46
35	0-49	RX0	CN2.3@c46
36	0-49	TX0	CN2.4@c46
37	0-49	GND	LED.K4@c10 LED.K1@c32 LED.K2@c36 LED.K3@c40 U2.GND@c44

Step 3 — every pin, where it goes

☐	Part.pin	Label on part	Net	Hole
☐	C1.1	1	GND	col 13, row 1
☐	C1.2	2	+3V3	col 13, row 2
☐	C2.1	1	GND	col 20, row 1
☐	C2.2	2	+3V3	col 20, row 2
☐	C5.1	1	GND	col 12, row 1
☐	C5.2	2	+5V	col 12, row 3
☐	CN1.1	1	+12V	col 2, row 0
☐	CN1.2	2	GND	col 2, row 1
☐	CN2.1	1	+5V	col 46, row 33
☐	CN2.2	2	GND	col 46, row 34
☐	CN2.3	3	RX0	col 46, row 35
☐	CN2.4	4	TX0	col 46, row 36
☐	LED.K1	K1	K1	col 32, row 29
☐	LED.K2	K2	K2	col 36, row 30
☐	LED.K3	K3	K3	col 40, row 31
☐	LED.K4	K4	K4	col 10, row 32
☐	LED.K4	K4	GND	col 10, row 37
☐	LED.K1	K1	GND	col 32, row 37
☐	LED.K2	K2	GND	col 36, row 37
☐	LED.K3	K3	GND	col 40, row 37
☐	R.K2	K2	LED_B	col 34, row 16
☐	R.K4	K4	LED_R	col 8, row 18
☐	R.K1	K1	LED_G	col 30, row 18
☐	R.K3	K3	LED_Y	col 38, row 19
☐	R.K1	K1	K1	col 30, row 29
☐	R.K2	K2	K2	col 34, row 30
☐	R.K3	K3	K3	col 38, row 31
☐	R.K4	K4	K4	col 8, row 32
—	U1A.1	3V3	LEAVE FREE	col 14, row 4
—	U1A.2	EN	LEAVE FREE	col 14, row 5
—	U1A.3	VP/I036	LEAVE FREE	col 14, row 6
—	U1A.4	VN/I039	LEAVE FREE	col 14, row 7
—	U1A.5	I034	LEAVE FREE	col 14, row 8
—	U1A.6	I035	LEAVE FREE	col 14, row 9
—	U1A.7	I032	LEAVE FREE	col 14, row 10
—	U1A.8	I033	LEAVE FREE	col 14, row 11
—	U1A.9	I025	LEAVE FREE	col 14, row 12

☐	U1A.10	I026	LORA_DIO0	col 14, row 13
—	U1A.11	I027	LEAVE FREE	col 14, row 14
☐	U1A.12	I014	LORA_RST	col 14, row 15
—	U1A.13	I012	LEAVE FREE	col 14, row 16
☐	U1A.14	GND	GND	col 14, row 17
☐	U1A.15	I013	LED_R	col 14, row 18
—	U1A.16	SD2	LEAVE FREE	col 14, row 19
—	U1A.17	SD3	LEAVE FREE	col 14, row 20
—	U1A.18	CMD	LEAVE FREE	col 14, row 21
☐	U1A.19	5V	+5V	col 14, row 22
☐	U1B.1	GND	GND	col 24, row 4
☐	U1B.2	I023	SPI_MOSI	col 24, row 5
☐	U1B.3	I022	I2C_SCL	col 24, row 6
☐	U1B.4	TX0	TX0	col 24, row 7
☐	U1B.5	RX0	RX0	col 24, row 8
☐	U1B.6	I021	I2C_SDA	col 24, row 9
☐	U1B.7	GND	GND	col 24, row 10
☐	U1B.8	I019	SPI_MISO	col 24, row 11
☐	U1B.9	I018	SPI_SCK	col 24, row 12
☐	U1B.10	I05	LORA_NSS	col 24, row 13
—	U1B.11	I017	LEAVE FREE	col 24, row 14
—	U1B.12	I016	LEAVE FREE	col 24, row 15
☐	U1B.13	I04	LED_B	col 24, row 16
—	U1B.14	I00	LEAVE FREE	col 24, row 17
☐	U1B.15	I02	LED_G	col 24, row 18
☐	U1B.16	I015	LED_Y	col 24, row 19
—	U1B.17	SD1	LEAVE FREE	col 24, row 20
—	U1B.18	SD0	LEAVE FREE	col 24, row 21
—	U1B.19	CLK	LEAVE FREE	col 24, row 22
☐	U2.VDD	VDD	+3V3	col 44, row 2
☐	U2.MOSI	MOSI	SPI_MOSI	col 44, row 5
☐	U2.MISO	MISO	SPI_MISO	col 44, row 11
☐	U2.SCK	SCK	SPI_SCK	col 44, row 12
☐	U2.NSS	NSS	LORA_NSS	col 44, row 13
☐	U2.RST	RST	LORA_RST	col 44, row 23
☐	U2.DIO0	DIO0	LORA_DIO0	col 44, row 24
☐	U2.GND	GND	GND	col 44, row 37
☐	U3.1	1	GND	col 6, row 25
☐	U3.2	2	+3V3	col 6, row 26
☐	U3.3	3	I2C_SCL	col 6, row 27

☐	U3.4	4	I2C_SDA	col 6, row 28
☐	U4.IN+	IN+	+12V	col 6, row 0
☐	U4.IN-	IN-	GND	col 6, row 1
☐	U4.OUT-	OUT-	GND	col 8, row 1
☐	U4.OUT+	OUT+	+5V	col 8, row 3
☐	U5.1	VIN	GND	col 11, row 1
☐	U5.2	GND	+3V3	col 11, row 2
☐	U5.3	SCL	+5V	col 11, row 3

Step 4 — solder 15 wire links

Insulated wire on the component side. Tick as you go, and buzz each one out after fitting.

☐	Net	From	To
☐	+3V3	col 11, row 2	col 6, row 26
☐	+5V	col 8, row 3	col 14, row 22
☐	+5V	col 14, row 22	col 46, row 33
☐	GND	col 2, row 1	col 24, row 4
☐	GND	col 6, row 25	col 46, row 34
☐	GND	col 14, row 17	col 6, row 25
☐	GND	col 24, row 4	col 24, row 10
☐	GND	col 24, row 10	col 14, row 17
☐	GND	col 46, row 34	col 10, row 37
☐	I2C_SCL	col 24, row 6	col 6, row 27
☐	I2C_SDA	col 24, row 9	col 6, row 28
☐	LORA_DIO0	col 14, row 13	col 44, row 24
☐	LORA_RST	col 14, row 15	col 44, row 23
☐	RX0	col 24, row 8	col 46, row 35
☐	TX0	col 24, row 7	col 46, row 36

Before power

☐	Meter check	Expect
☐	GND to +3V3	open
☐	GND to +5V	open
☐	GND to VBAT / +12V	open
☐	every cut, across it	open